

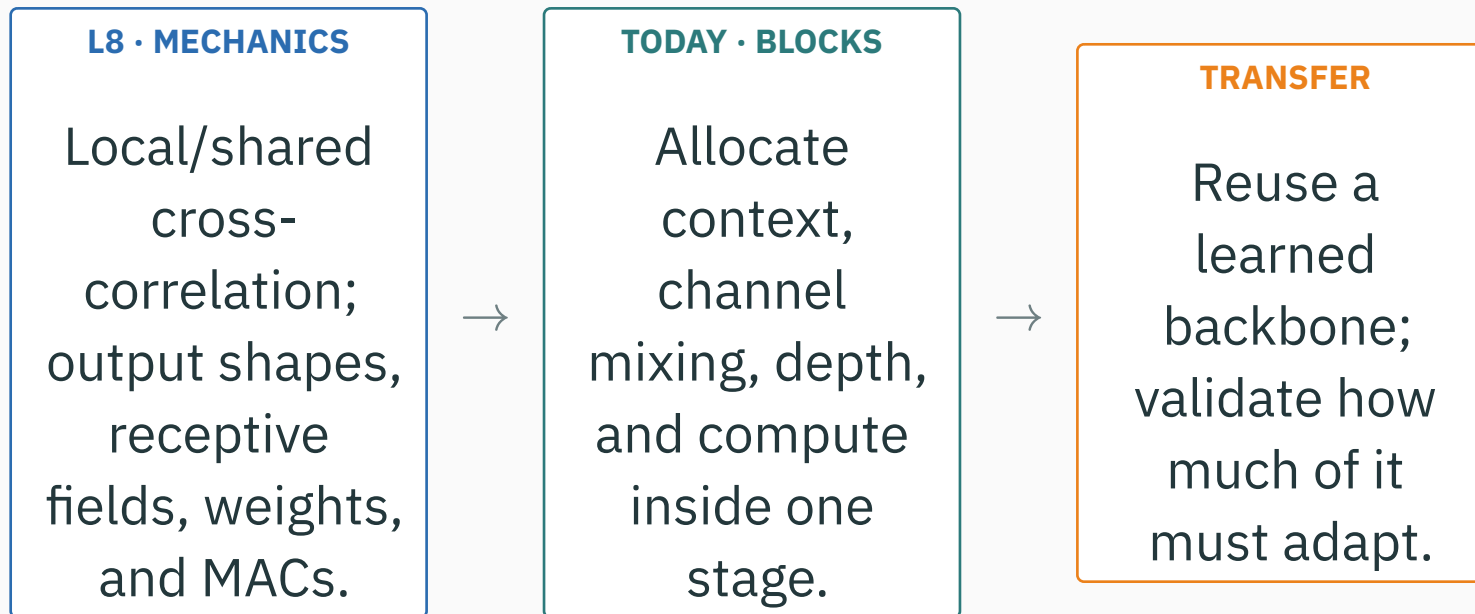
Modern CNN Pipelines and Transfer Learning

Blocks, Scaling, Residuals, and Reusing Backbones

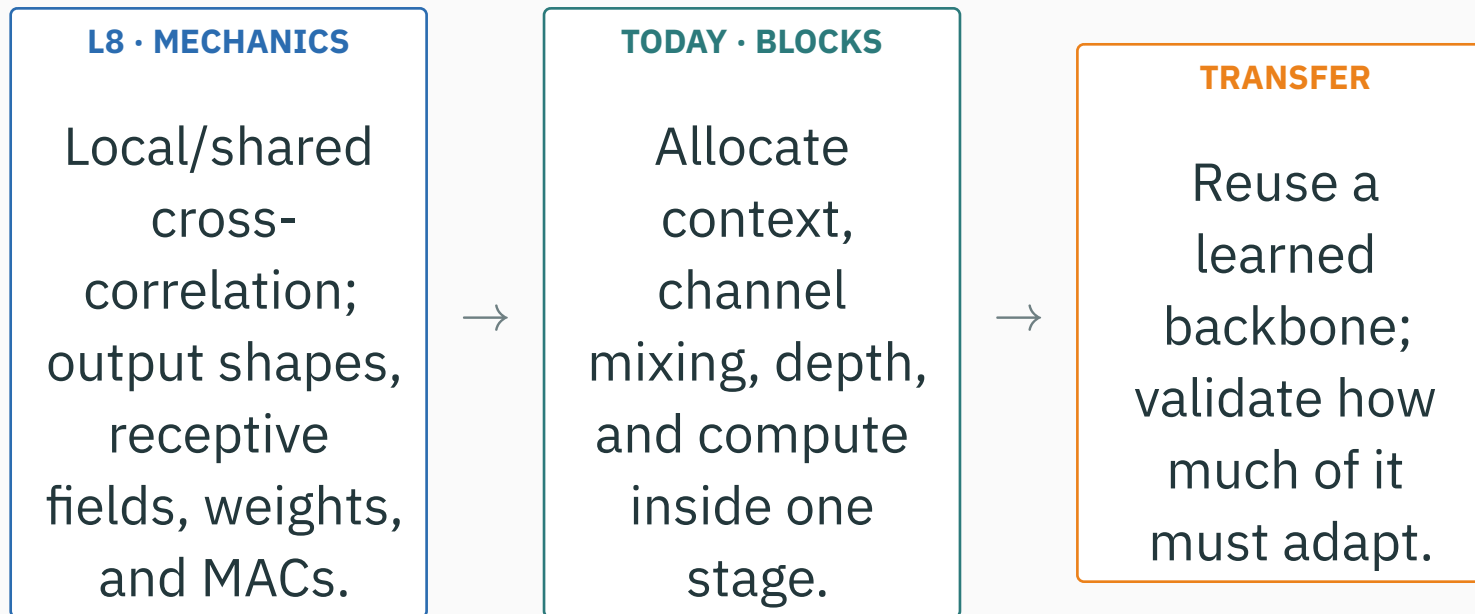
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ES 667 · Deep Learning · IIT Gandhinagar

L8 gave us the operator; today we spend it deliberately



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L6 already established why residual routes help optimization. L7 fixed the validation/test protocol. We will retrieve both—then use them inside a deployable CNN pipeline.

Commit: which design can meet the stage budget?

QUESTION

HELD CASE · DEPLOYMENT CONTRACT $28 \times 28 \times 64 \rightarrow 28 \times 28 \times 128$, same spatial grid, receptive field at least 3×3 , fewer than 10 million convolution MACs.

A · STANDARD

one shared 3×3 , $64 \rightarrow 128$

B · FACTORIZED

separate spatial filtering
from channel mixing

C · EITHER

sharing alone guarantees
the budget

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Commit to A, B, or C. Write the count you would calculate first; keep the choice until the closing revisit.

The student contract and 80-minute route

By the end, you should be able to turn a block diagram into a shape/compute ledger, justify one design move, and run a leak-free probe-first transfer experiment.

CORE · 55 MIN

bridge → context +
bottleneck → residuals →
separable convolution →
transfer → revisit

SHOULD · 15 MIN

Inception branches →
frozen state +
preprocessing →
diagnostic tests

OPTIONAL · 10 MIN

groups + scaling +
ConvNeXt → FPN teaser
→ primary provenance

— held input — learned operation — context / shape — compute / adaptation — verified — failure

One ledger will carry every design decision

HELD CASE · LEDGER CONVENTION All stage candidates use $H_{\text{out}} = W_{\text{out}} = 28$, stride 1, stated same padding, and no bias. One MAC means one multiply-accumulate per weight per output location.

candidate	output	RF	weights	MACs
standard 3×3	$28 \times 28 \times$ 128	3×3	?	?

Budget: fewer than 10,000,000 MACs. Shape alone is not enough.

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Every candidate must preserve the output contract; only context, factorization, route, and cost may change.

Baseline: count the standard 3×3 before optimizing

QUESTION

HELD CASE · CANDIDATE 0 One same-padded 3×3 convolution maps $64 \rightarrow 128$ channels at 28×28 output locations. Bias is omitted.

$$\text{weights} = 3 \cdot 3 \cdot 64 \cdot 128$$

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Predict both exact counts and whether the design meets the 10M budget.

Baseline answer: sharing is necessary but not sufficient

A ANSWER

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A ANSWER

candidate	output	RF	weights	MACs
standard 3×3	$28 \times 28 \times 128$	3×3	73, 728	57, 802, 752

$57.803\text{M} > 10\text{M}$: the output contract passes; the deployment budget fails.

WHAT L8 BOUGHT

One kernel bank is reused at all 784 locations.

WHAT REMAINS

Reduce work inside the kernel bank without changing the output.

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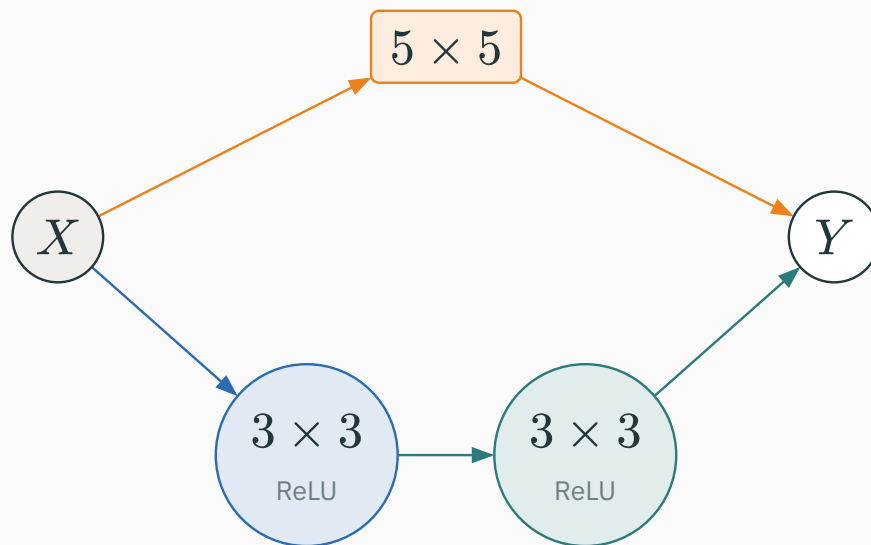
Reduce work inside the kernel bank without changing the output.

The lecture is now a controlled search for a cheaper stage—not a tour of architecture names.

Design move 1 — grow context with small kernels

HELD CASE · CONTEXT Keep the held $28 \times 28 \times 64 \rightarrow 28 \times 28 \times 128$ output. Now require a 5×5 receptive field.

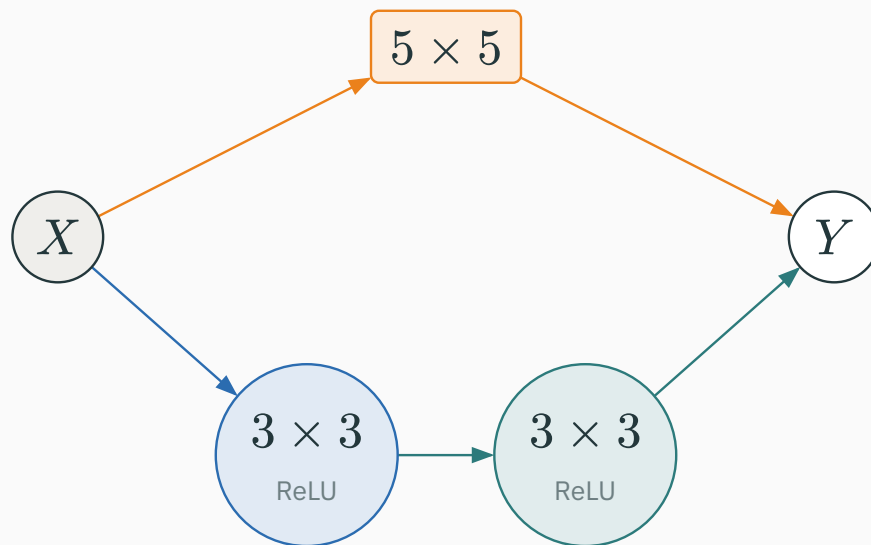
Should we use one 5×5 convolution—or compose two same-padded 3×3 s?



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Both paths see the same spatial extent. Only the blue–teal path inserts an extra learned nonlinearity.

Receptive field: derive it one layer at a time

D DERIVATION

With stride 1, each $k \times k$ layer grows the receptive-field side length by $k - 1$:

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$$r_\ell = r_{\ell-1} + (k - 1)$$
$$1 \xrightarrow{3 \times 3} 3 \xrightarrow{3 \times 3} 5 \xrightarrow{3 \times 3} 7 \xrightarrow{3 \times 3} 9$$

Four stacked 3×3 layers match one 9×9 receptive field.

They also insert **four** transformations and **four** activations instead of one.

Depth grows context gradually—and learns what to keep after each step.

Same output and context: count both candidates

The stacked path uses $64 \rightarrow 64 \rightarrow 128$; the direct path uses $64 \rightarrow 128$. Biases are omitted.

	two 3×3s	one 5×5
weights	$9(64 \cdot 64 + 64 \cdot 128)$	$25 \cdot 64 \cdot 128$
count	110, 592	204, 800
ReLUs	2	1

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count	110, 592	204, 800
ReLU	2	1

$204,800 - 110,592 = 94,208$ fewer weights: a 46% reduction.

Small kernels are not merely aesthetic: they buy equal context, fewer weights, and more nonlinear stages.

Ledger: more context still misses the deployment budget

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candidate	output	RF	weights	MACs
standard 3×3	$28 \times 28 \times 128$	3×3	73,728	57.803M
two 3×3 s	$28 \times 28 \times 128$	5×5	110,592	86.704M
one 5×5	$28 \times 28 \times 128$	5×5	204,800	160.563 M

All three preserve shape with stated same padding. None meets 10M.

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All three preserve shape with stated same padding. None meets 10M.

Stacking small kernels is a context decision. It does not, by itself, solve the channel-pair cost.

Misconception check: does depth always grow spatial context?

QUESTION

Stack ten 1×1 convolutions. What is the receptive field?

A · 1×1

B · 10×10

C · depends on channels

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Stack ten 1×1 convolutions. What is the receptive field?

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Answer: A. Depth composes transformations; kernel size and stride determine spatial reach.

Design move 2 — mix channels without mixing neighbours

At one spatial location, collect the C_{in} channel values into a vector x_{hw} :

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At one spatial location, collect the C_{in} channel values into a vector x_{hw} :

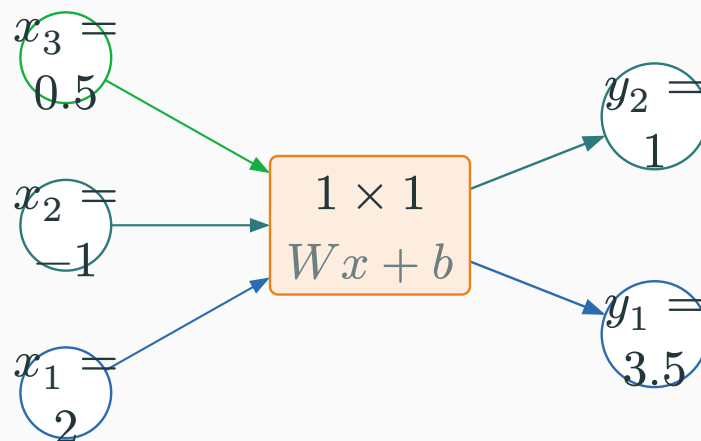
$$y_{hw} = Wx_{hw} + b, \quad W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$$

The **same** matrix W is used at every location.

No value from $(h + 1, w)$ or $(h, w + 1)$ is read.

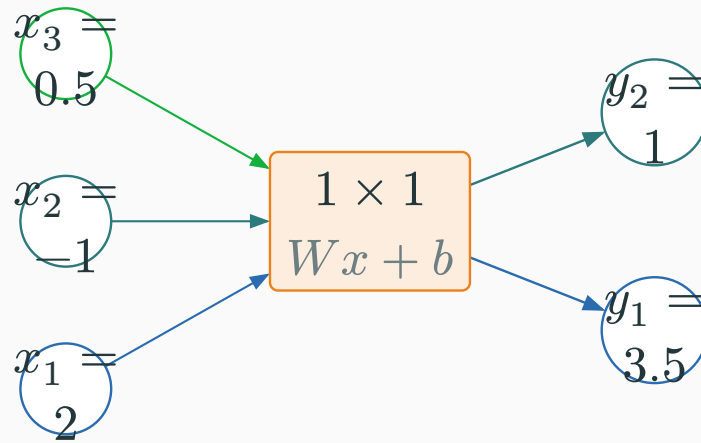
A 1×1 convolution is a shared, per-pixel linear layer across channels.

One pixel, two learned channel combinations



For $x = (2, -1, 0.5)^\top$, choose $W = \begin{pmatrix} 1 & -1 & 0 \\ 0.5 & 0 & 2 \end{pmatrix}$ and $b = (0.5, -1)^\top$.

One pixel, two learned channel combinations



For $x = (2, -1, 0.5)^\top$, choose $W = \begin{pmatrix} 1 & -1 & 0 \\ 0.5 & 0 & 2 \end{pmatrix}$ and $b = (0.5, -1)^\top$.

$$y = \begin{pmatrix} 1 & -1 & 0 \\ 0.5 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 0.5 \end{pmatrix} + \begin{pmatrix} 0.5 \\ -1 \end{pmatrix} = \begin{pmatrix} 3.5 \\ 1 \end{pmatrix}.$$

Two output channels = two different learned summaries of the same input channels.

The same channel mixer runs over the whole grid

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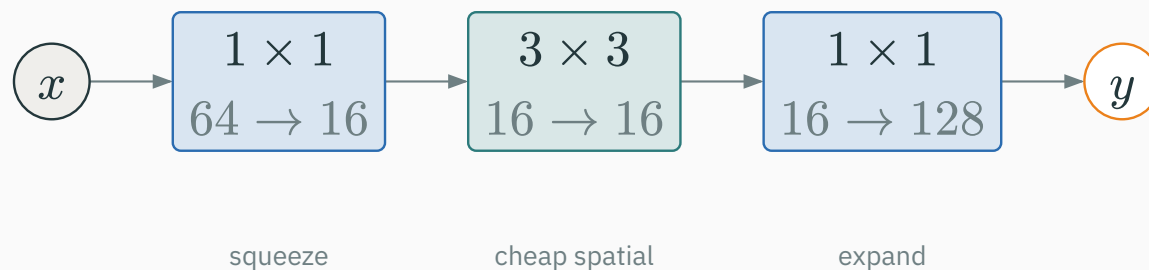
x_{hw}	$(0, 0)$	$(0, 1)$	$(1, 0)$	$(1, 1)$
three channels	$\begin{pmatrix} 2 \\ -1 \\ 0.5 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$	$\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$
$Wx + b$	$\begin{pmatrix} 3.5 \\ 1 \end{pmatrix}$	$\begin{pmatrix} -0.5 \\ 3 \end{pmatrix}$	$\begin{pmatrix} -1.5 \\ -1.5 \end{pmatrix}$	$\begin{pmatrix} 3.5 \\ 2.5 \end{pmatrix}$

$$2 \times 2 \times 3 \xrightarrow{\text{same } 1 \times 1 \text{ weights}} 2 \times 2 \times 2$$

Spatial coordinates stay aligned; only the channel representation changes.

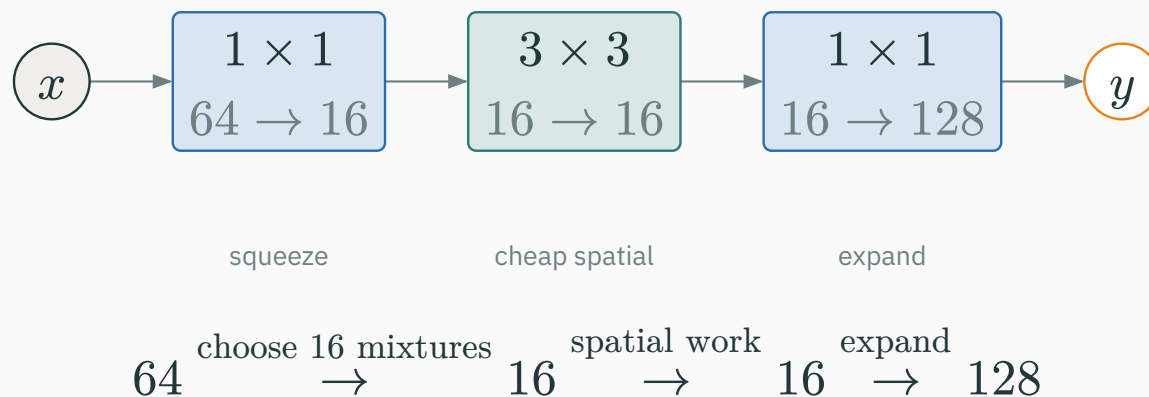
Bottleneck: spend the 3×3 in a narrow channel space

HELD CASE · CANDIDATE 2 Narrow $64 \rightarrow 16$, perform the spatial work at width 16, then expand $16 \rightarrow 128$. Same 28×28 grid throughout.



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HELD CASE · CANDIDATE 2 Narrow $64 \rightarrow 16$, perform the spatial work at width 16, then expand $16 \rightarrow 128$. Same 28×28 grid throughout.



The two 1×1 s learn which channel combinations deserve expensive spatial processing.

For the held 28×28 stage, compare the direct $3 \times 3, 64 \rightarrow 128$, with $64 \rightarrow 16 \rightarrow 16 \rightarrow 128$:

direct weights $= 3 \cdot 3 \cdot 64 \cdot 128 = 73,728$.

With the bottleneck:

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With the bottleneck:

$$64 \cdot 16 + 3 \cdot 3 \cdot 16 \cdot 16 + 16 \cdot 128 = 1,024 + 2,304 + 2,048 = 5,376.$$

Each weight is reused at $28^2 = 784$ locations: 57.803 million versus 4.215 million MACs.

$5,376 \cdot 784 = 4,214,784$: the first candidate under the 10M budget.

Misconception check: what did the 1×1 save?

A ANSWER

It did **not** make the final 3×3 kernel spatially smaller.

wrong: $3 \times 3 \rightarrow 1 \times 1$ spatial window

right: $64 \rightarrow 16$ channels **before** the 3×3

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The saving comes from fewer input–output channel pairs inside the expensive spatial convolution.

Ledger: channel compression changes the decision

candidate	output	RF	weights	MACs
standard 3×3	$28 \times 28 \times 128$	3×3	73,728	57.803M
two 3×3 s	$28 \times 28 \times 128$	5×5	110,592	86.704M
bottleneck 16	$28 \times 28 \times 128$	3×3	5,376	4.215M

GREEN means the candidate satisfies shape, receptive-field, and compute contracts.

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GREEN means the candidate satisfies shape, receptive-field, and compute contracts.

The bottleneck clears the budget with 5.785M MACs to spare.

SHOULD · let one block inspect several scales

HELD CASE · CANDIDATE 3 The held stage may need channel evidence, 3×3 texture, 5×5 context, and a pooled summary at the same location.

Can four branches preserve the 28×28 grid, concatenate to 128 channels, and stay under budget?

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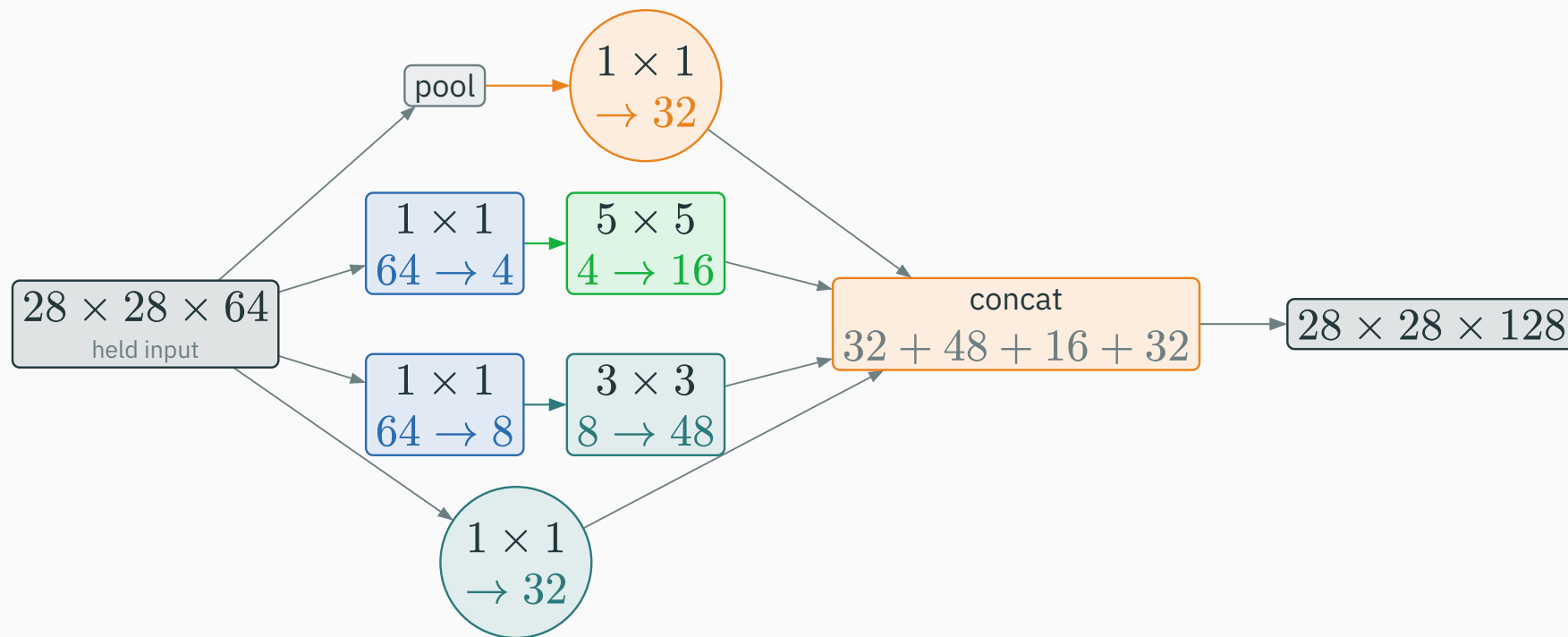
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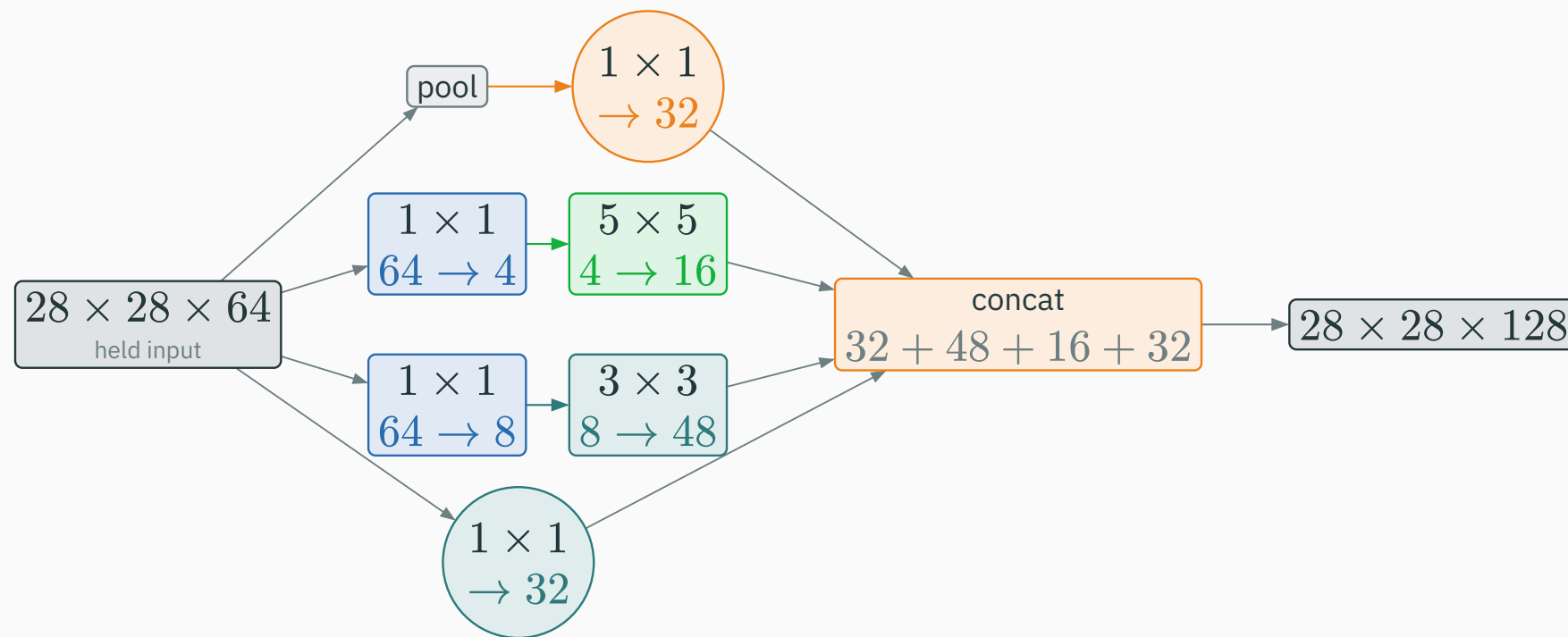
1×1 channel evidence 3×3 local pattern 5×5 wider context pooling summary

Inception runs alternatives in parallel; the stage output keeps every branch as a channel group.

A shaped Inception candidate: every width is visible



A shaped Inception candidate: every width is visible



$32 + 48 + 16 + 32 = 128$ output channels; height and width remain 28×28 .

Concatenation stacks channels; it does not average or add branch outputs.

Count the multi-scale candidate branch by branch

D DERIVATION

All convolutions use stride 1. The 3×3 , 5×5 , and 3×3 pool use padding 1, 2, and 1, so every branch remains 28×28 .

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All convolutions use stride 1. The 3×3 , 5×5 , and 3×3 pool use padding 1, 2, and 1, so every branch remains 28×28 .

branch	weight expression	weights
$1 \times 1 \rightarrow 32$	$64 \cdot 32$	2,048
$1 \times 1 \rightarrow$ $8 \xrightarrow{3 \times 3} 48$	$64 \cdot 8 + 9 \cdot 8 \cdot 48$	3,968
$1 \times 1 \rightarrow$ $4 \xrightarrow{5 \times 5} 16$	$64 \cdot 4 + 25 \cdot 4 \cdot 16$	1,856
pool $\xrightarrow{1 \times 1} 32$	$64 \cdot 32$	2,048

9,920 weights and $9,920 \cdot 784 = 7,777,280$ convolution MACs. Pooling comparisons are excluded from this ledger.

Concatenate only after every spatial shape agrees

D DERIVATION

$28 \times 28 \times 32; \parallel; 28 \times 28 \times 48; \parallel; 28 \times 28 \times 16; \parallel; 28 \times 28 \times 32$

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$28 \times 28 \times 32; \parallel; 28 \times 28 \times 48; \parallel; 28 \times 28 \times 16; \parallel; 28 \times 28 \times 32$

concatenate $\rightarrow 28 \times 28 \times (32 + 48 + 16 + 32) = 28 \times 28 \times 128$

The candidate meets shape and compute contracts; its widest branch sees a 5×5 neighbourhood.

Which statement is true?

A. Every signal crosses all four branches in sequence.

B. Each branch offers an alternative transform at one stage; concatenation preserves all outputs.

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Answer: B. Branching widens the choice of scale; stacking modules supplies depth.

L6 retrieval · learn a correction, not a replacement

Suppose a useful block should mostly preserve its input. A plain block must learn $H(x) \approx x$.

L6 retrieval · learn a correction, not a replacement

Suppose a useful block should mostly preserve its input. A plain block must learn $H(x) \approx x$.

A residual block instead writes

$$H(x) = x + F(x)$$

If “do nothing” is best, the learned branch only needs $F(x) \approx 0$.

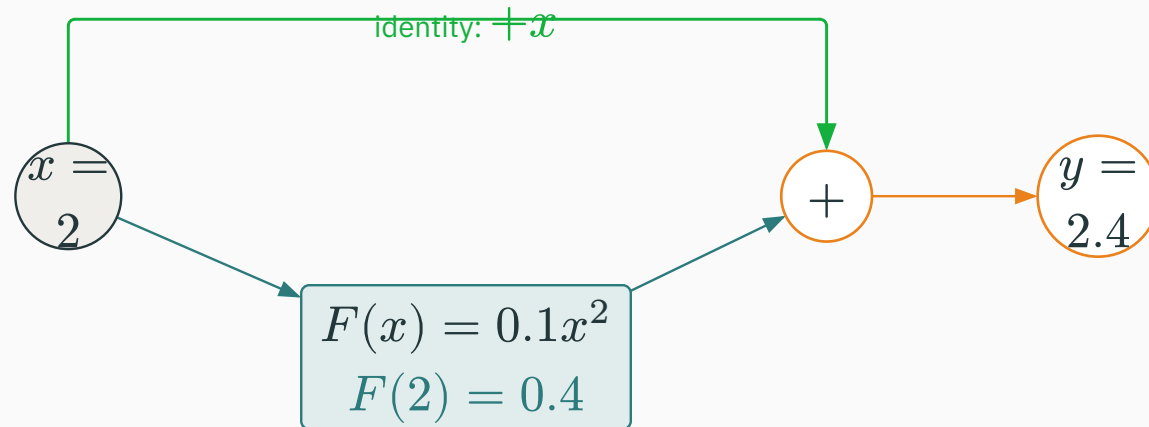
The shortcut makes identity behaviour explicit; the residual path learns only the needed change.

HELD CASE · ONE SCALAR BLOCK At $x = 2$, let $F(x) = 0.1x^2$. Predict $F(2)$, $y = x + F(x)$, and $d\frac{y}{d}x$.

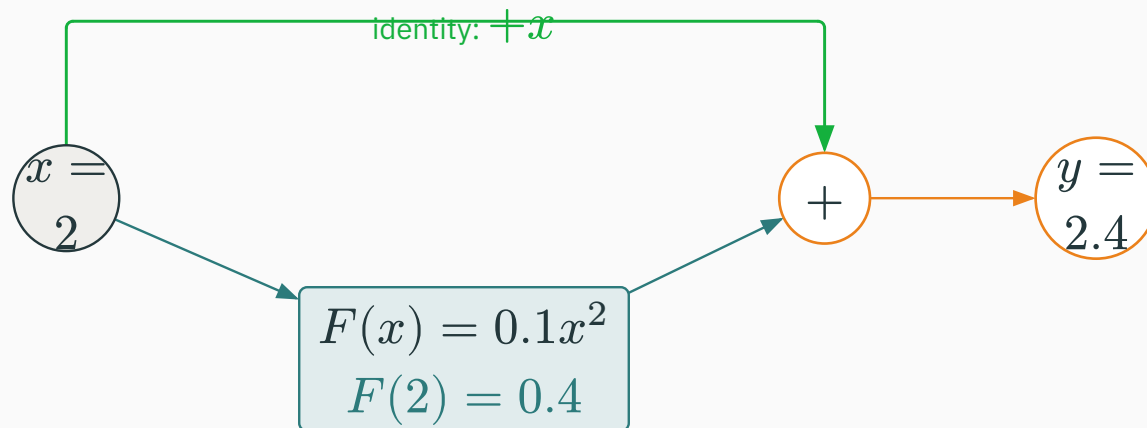
Commit: residual arithmetic on one scalar

QUESTION

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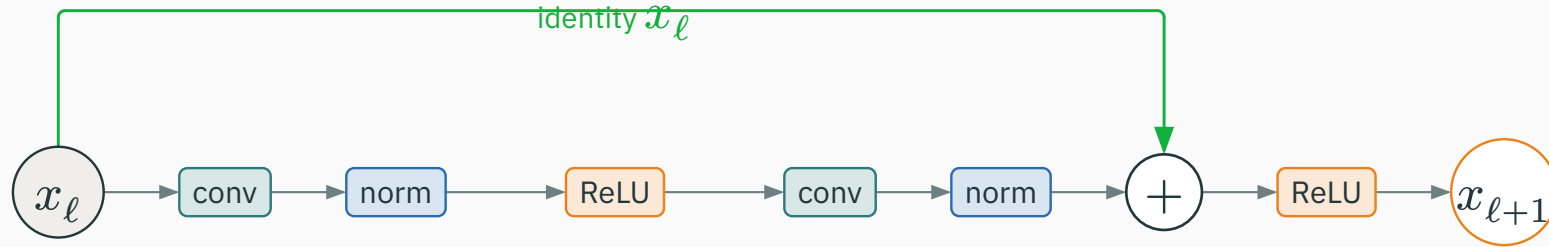
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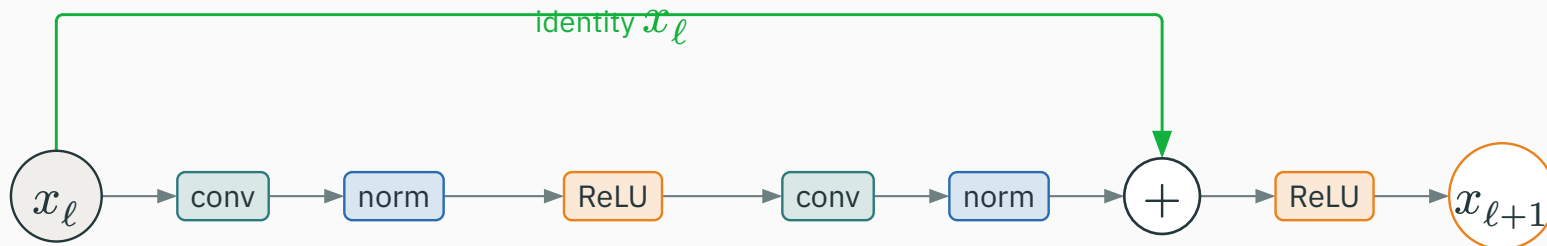
$$F(2) = 0.1(2)^2 = 0.4, \quad y = x + F(x) = 2 + 0.4 = 2.4.$$

The output keeps the original 2 and adds the learned correction 0.4.

The residual block as a node diagram



The residual block as a node diagram



$F(x_\ell)$ travels through learned layers; x_ℓ travels unchanged along the green edge.

The addition node is only legal when the two incoming tensors have identical shapes.

Why the gradient has a direct route

For $y = x + F(x)$,

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$$\frac{\partial y}{\partial x} = 1 + \frac{\partial F}{\partial x}.$$

Continue the scalar example $F(x) = 0.1x^2$ at $x = 2$:

$F'(x) = 0.2x$, so $F'(2) = 0.4$ and

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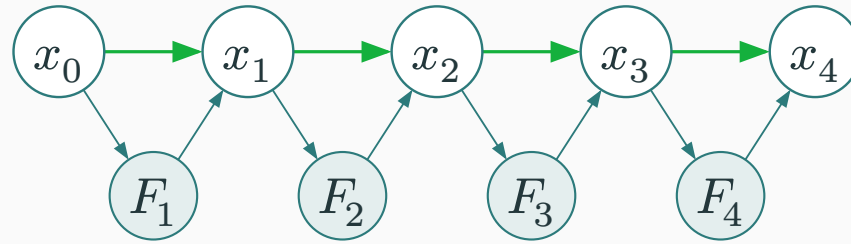
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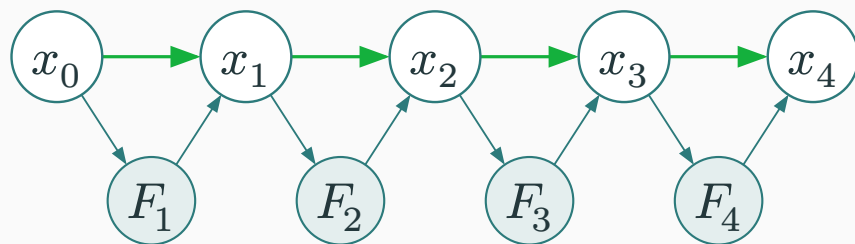
$F'(x) = 0.2x$, so $F'(2) = 0.4$ and

$$\frac{dy}{dx} = 1 + 0.4 = 1.4.$$

The 1 is a direct derivative term. It helps signal flow; it does not guarantee a large total gradient if other terms cancel.

Residual depth: the identity path survives many blocks





$$x_4 = x_0 + F_1(x_0) + F_2(x_1) + F_3(x_2) + F_4(x_3)$$

Depth becomes a sequence of refinements around a persistent representation.

The held channel change makes an identity shortcut illegal

QUESTION

HELD CASE · CANDIDATE 4 $x \in \mathbb{R}^{28 \times 28 \times 64}$ but the required stage output is $28 \times 28 \times 128$. Can $x + F(x)$ be added directly?

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$$F(x) : 28 \times 28 \times 64 \rightarrow 28 \times 28 \times 128.$$

The identity tensor cannot be added as-is. Apply a 1×1 , stride-1 projection:

$$P(x) : 28 \times 28 \times 64 \rightarrow 28 \times 28 \times 128.$$

Now the addition is well-defined:

$$y = F(x) + P(x).$$

Projection shortcuts change shape; identity shortcuts preserve it.

A projected residual candidate still clears the budget

D DERIVATION

Use $F : 64 \rightarrow 8 \rightarrow 8 \rightarrow 128$ with $1 \times 1, 3 \times 3, 1 \times 1$; use $P : 64 \rightarrow 128$ with 1×1 . Biases are omitted.

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Use $F : 64 \rightarrow 8 \rightarrow 8 \rightarrow 128$ with $1 \times 1, 3 \times 3, 1 \times 1$; use $P : 64 \rightarrow 128$ with 1×1 . Biases are omitted.

$$\text{weights}_F = 64 \cdot 8 + 9 \cdot 8 \cdot 8 + 8 \cdot 128 = 2,112,$$

$$\text{weights}_P = 64 \cdot 128 = 8,192.$$

A projected residual candidate still clears the budget

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Use $F : 64 \rightarrow 8 \rightarrow 8 \rightarrow 128$ with $1 \times 1, 3 \times 3, 1 \times 1$; use $P : 64 \rightarrow 128$ with 1×1 . Biases are omitted.

$$\text{weights}_F = 64 \cdot 8 + 9 \cdot 8 \cdot 8 + 8 \cdot 128 = 2,112,$$

$$\text{weights}_P = 64 \cdot 128 = 8,192.$$

candidate	output	RF	weights	MACs
projected residual	$28 \times 28 \times 128$	3×3	10,304	8,078,336

$y = F(x) + P(x)$ is legal; both routes end at the same shape.

The shortcut is not free when shape changes—but this narrow residual branch remains within budget.

Misconception check: is the shortcut “skipping learning”?

A ANSWER

No. The output still depends on the learned residual $F(x)$.

$$x + F(x) = y$$

preserved representation + learned update

The shortcut changes the **parameterization** of the target function. It makes preservation cheap while leaving the branch free to learn large or small changes.

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$$x + F(x) = y$$

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The shortcut changes the **parameterization** of the target function. It makes preservation cheap while leaving the branch free to learn large or small changes.

Residual learning is controlled refinement—not a bypass around the model.

Core move · factor spatial filtering from channel mixing

A standard $k \times k$ convolution does both jobs at once:

Core move · factor spatial filtering from channel mixing

A standard $k \times k$ convolution does both jobs at once:

$$k^2 \cdot C_{\text{in}} C_{\text{out}}$$

$$k^2 \text{ spatial positions} \quad \times \quad C_{\text{in}} C_{\text{out}} \text{ channel pairs}$$

Ignoring bias, its weight count is $k^2 C_{\text{in}} C_{\text{out}}$; its MAC count is

$$H_{\text{out}} W_{\text{out}} k^2 C_{\text{in}} C_{\text{out}}.$$

Depthwise separable convolution performs the two jobs in two explicit steps.

Standard convolution: establish the baseline

Let $H = W = 28$, $k = 3$, $C_{\text{in}} = 64$, and $C_{\text{out}} = 128$.

Standard convolution: establish the baseline

Let $H = W = 28$, $k = 3$, $C_{\text{in}} = 64$, and $C_{\text{out}} = 128$.

$$\text{weights} = 3 \cdot 3 \cdot 64 \cdot 128 = 73,728.$$

There are $28 \cdot 28 = 784$ output locations:

$$\text{MACs} = 784 \cdot 73,728 = 57,802,752.$$

This is the cost the factorized version must match in output shape—not in arithmetic.

Depthwise step: one spatial filter per input channel

D DERIVATION

$$28 \times 28 \times 64 \xrightarrow{64 \text{ separate } 3 \times 3 \text{ filters}} 28 \times 28 \times 64$$

Depthwise step: one spatial filter per input channel

D DERIVATION

$$28 \times 28 \times 64 \xrightarrow{64 \text{ separate } 3 \times 3 \text{ filters}} 28 \times 28 \times 64$$

$$\text{weights} = 3 \cdot 3 \cdot 64 = 576,$$

$$\text{MACs} = 784 \cdot 576 = 451,584.$$

No channel is mixed with another yet.

Depthwise convolution answers only: “what spatial pattern occurs within this channel?”

Pointwise step: mix the filtered channels

D DERIVATION

$$28 \times 28 \times 64 \xrightarrow{1 \times 1, 128 \text{ outputs}} 28 \times 28 \times 128$$

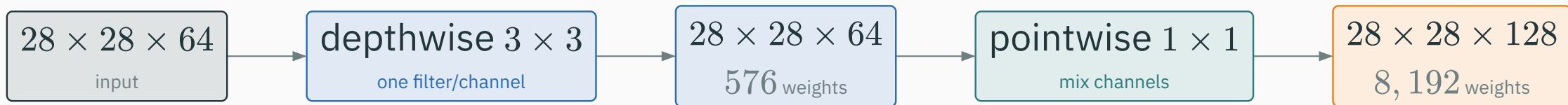
$$28 \times 28 \times 64 \xrightarrow{1 \times 1, 128 \text{ outputs}} 28 \times 28 \times 128$$

$$\text{weights} = 64 \cdot 128 = 8,192,$$

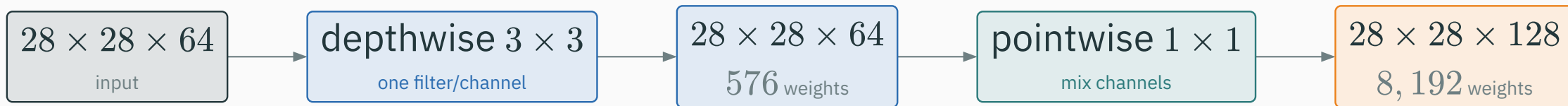
$$\text{MACs} = 784 \cdot 8,192 = 6,422,528.$$

The pointwise layer answers: “which channel combinations should form the 128 outputs?”

Put the factorized block back together



Put the factorized block back together



$$\text{separable weights} = 576 + 8,192 = 8,768,$$

$$\text{separable MACs} = 451,584 + 6,422,528 = 6,874,112.$$

$$\frac{57,802,752}{6,874,112} \approx 8.41 \times \text{fewer MACs than the standard convolution.}$$

one group $\rightarrow$ a standard convolution mixes **all** input channels

g groups $\rightarrow$ each output sees only $\frac{C_{\text{in}}}{g}$ input channels

$g = C_{\text{in}} \rightarrow$ depthwise connectivity: each output sees one input channel

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$g = C_{\text{in}} \rightarrow$ depthwise connectivity: each output sees one input channel

With depth multiplier 1, $C_{\text{out}} = C_{\text{in}}$ and there is one spatial filter per channel; a later 1×1 restores full mixing.

For a representative convolution, MACs scale approximately as

$$\text{MACs} \propto r^2 d w^2,$$

where r scales resolution, d depth, and w channel width.

Double	What changes	rough MAC multiplier
resolution r	$H, W \rightarrow 2H, 2W$	$4 \times$
depth d	twice as many blocks	$2 \times$
width w	$C_{\text{in}}, C_{\text{out}} \rightarrow 2 \times$	$4 \times$

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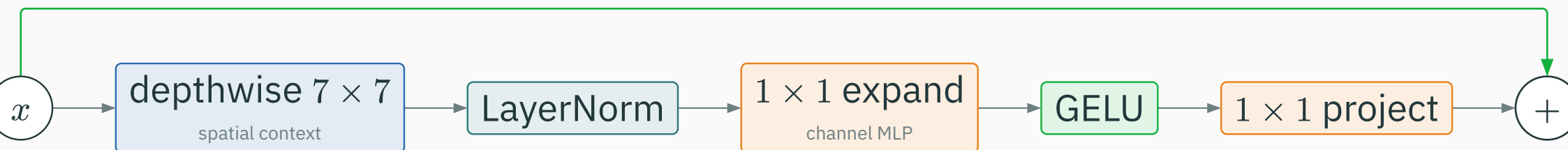
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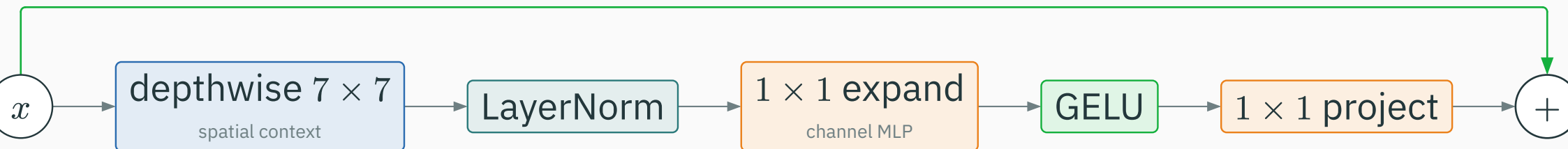
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EfficientNet's compound scaling grows all three gradually instead of overspending on one axis.

OPTIONAL · ConvNeXt recombines familiar moves

V VISUAL





Modern CNNs recombine the same ideas: large-context depthwise filtering, cheap channel mixing, normalization, and a residual path.

Does $8.41 \times$ fewer MACs guarantee $8.41 \times$ lower wall-clock latency?

Does $8.41 \times$ fewer MACs guarantee $8.41 \times$ lower wall-clock latency?

Not necessarily.

Wall-clock speed also depends on memory traffic, kernel launch overhead, tensor layout, batch size, and hardware support.

algorithmic cost $\neq$ measured latency

Use MACs and parameters for reasoning; benchmark latency on the deployment device before making the final choice.

The stage ledger now contains architectural choices

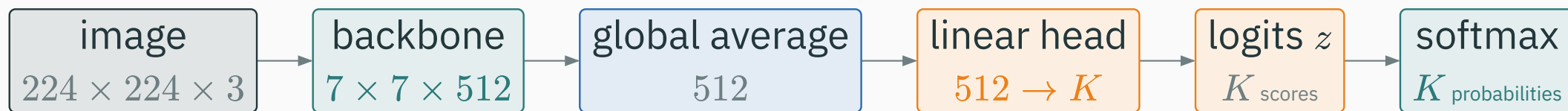
candidate	output	RF	weights	MACs
standard 3×3	$28^2 \times 128$	3×3	73,728	57.803M
two 3×3 s	$28^2 \times 128$	5×5	110,592	86.704M
bottleneck 16	$28^2 \times 128$	3×3	5,376	4.215M
Inception	$28^2 \times 128$	up to 5×5	9,920	7.777M
projected residual	$28^2 \times 128$	3×3	10,304	8.078M
depthwise + pointwise	$28^2 \times 128$	3×3	8,768	6.874M

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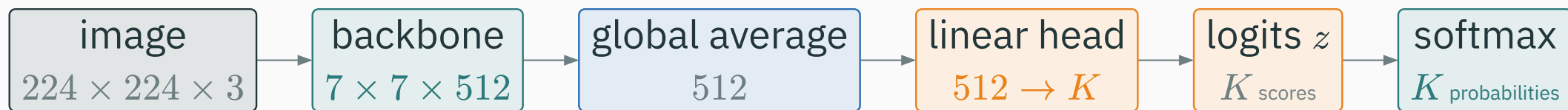
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The best block is conditional: satisfy the task contract, then measure the resource that matters on the target hardware.

A stage becomes one repeated part of a backbone



A stage becomes one repeated part of a backbone



STAGES

POOL

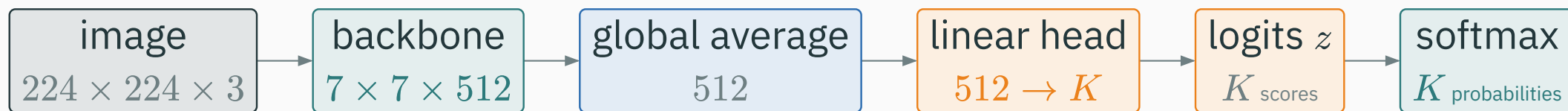
HEAD

spatial transforms build a hierarchy

$$7 \times 7 \times 512 \rightarrow h \in \mathbb{R}^{512}$$

$$h \rightarrow z \in \mathbb{R}^K$$

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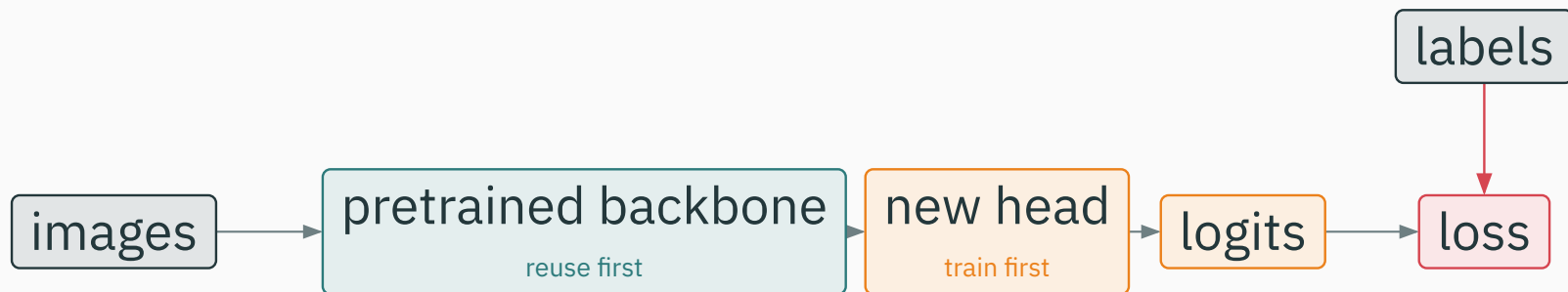
The block ledger governs the backbone; transfer asks how much of that learned hierarchy to reuse.

Transfer starts with a hypothesis: old features may already separate new labels

pixels → edges and colours → textures → parts → task evidence

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pixels → edges and colours → textures → parts → task evidence



Probe the reused representation first; adapt it only when validation evidence says the fixed features are insufficient.

Commit: probe, unfreeze late, or fine-tune all?

QUESTION

A pretrained backbone emits $h \in \mathbb{R}^{512}$ for a new six-class dataset. Training data are limited; the test set is sealed.

A · PROBE

freeze the backbone;
train only $512 \rightarrow 6$

B · LATE BLOCKS

train the head and final
backbone stage

C · ALL

fine-tune every
parameter immediately

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Commit before seeing target validation curves. The correct choice is empirical, not a slogan.

Count the new six-class head exactly

D DERIVATION

$$z = Wh + b, \quad W \in \mathbb{R}^{6 \times 512}, \quad b \in \mathbb{R}^6.$$

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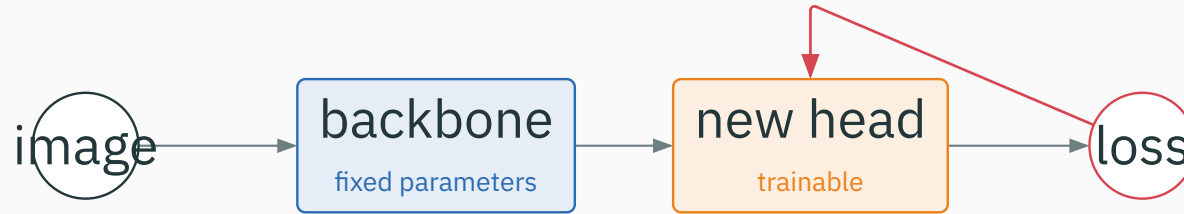
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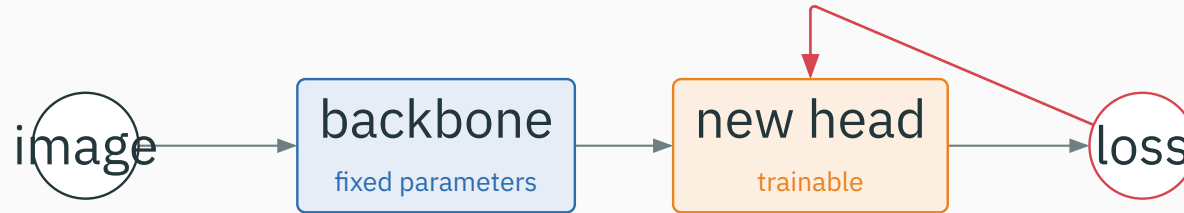
$$\text{trainable head parameters} = 6 \cdot 512 + 6 = 3,078.$$

If the backbone has 11 million parameters, a linear probe trains $\frac{3,078}{11,000,000 + 3,078} \approx 0.028\%$ of the model.

The first transfer experiment is intentionally tiny and inspectable.

A linear probe computes through the backbone but updates only the head V VISUAL





A probe measures linear separability in pretrained feature space; it does not make backbone computation disappear.

SHOULD · freezing parameters does not freeze model state

D DERIVATION

PARAMETERS

`requires_grad=False`;
exclude them from the
optimizer.

BUFFERS

BatchNorm running
means/variances can still
change in `train()` mode.

STOCHASTIC LAYERS

Dropout also changes
behaviour between
`train()` and `eval()`.

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For a truly fixed probe, keep the frozen backbone in `eval()` while the new head trains. Updating BatchNorm statistics deliberately is a different intervention—name and validate it.

SHOULD · match both the preprocessing and tensor-layout contracts

D DERIVATION

BOARD NOTATION

$H \times W \times C$ keeps the spatial story readable.

PYTORCH INPUT

Conv2d receives $N \times C \times H \times W$.

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Channel order, input range, resize/crop policy, and (μ, σ) must match the weight recipe.

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Before blaming transfer, inspect one batch: shape, dtype, range, channel order, normalization, and backbone mode.

L7 protocol: validation chooses how much to unfreeze

TRAIN

fit head/backbone
parameters for one
declared regime

VALIDATION

choose probe vs late/all,
learning rates, stopping
epoch, and checkpoint

SEALED TEST

run once after the full
decision rule and
checkpoint are frozen

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**Repeatedly checking test performance while choosing unfreeze depth
turns the test set into validation data.**

A controlled transfer experiment changes one lever at a time

D DERIVATION

Hold fixed the seed, train/validation split, head initialization, augmentation, optimizer family, epoch budget, and selection metric.

A controlled transfer experiment changes one lever at a time

Hold fixed the seed, train/validation split, head initialization, augmentation, optimizer family, epoch budget, and selection metric.

regime	trainable blocks	backbone LR	record
probe	head only	0	best validation; epoch
late	head + final stage	small	same record
all	entire network	smaller still	same record

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Validation compares interventions fairly; the sealed test evaluates only the selected procedure.

Unfreeze progressively in response to evidence

probe underfits train and validation → reused features may not separate the task

unfreeze late blocks → test whether task-specific adaptation closes both gaps

consider broader unfreezing → only as another validation-controlled candidate

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High training accuracy but poor validation is not evidence to unfreeze more. It points first to data size, regularization, augmentation, or shift.

Parameter groups make the fine-tuning hypothesis explicit

D DERIVATION

$$\theta_h \xrightarrow{\eta_h} \theta_h - \eta_h \nabla_{\theta_h} \mathcal{L}, \quad \theta_b \xrightarrow{\eta_b} \theta_b - \eta_b \nabla_{\theta_b} \mathcal{L}$$

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$$0 < \eta_b < \eta_h$$

NEW HEAD

larger step: it starts random and must learn the task

REUSED BACKBONE

smaller step: preserve useful features while adapting

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smaller step: preserve useful features while adapting

The ratio is a hyperparameter selected on validation—not a universal constant.

SHOULD · diagnose transfer failures as symptom → suspect → test

symptom	suspect	smallest discriminating test
loss stays near random	input contract mismatch	log batch shape/range; compare expected transform
frozen features drift	BN buffers or dropout active	diff parameters and buffers across one step
validation drops after unfreeze	backbone rate too large	reduce η_b ; log layer-wise update norms
MACs fall; latency does not	memory/kernel overhead	benchmark the target device and batch size

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Change the experiment only after the test separates competing causes.

Follow along: predict, calculate, run, explain

I INTERACTIVE

INTERACTIVE

↗ colab.research.google.com/github/nipunbatra/dl-teaching/blob/master/notebooks/L08B/01_depthwise_separable_and_transfer_head.ipynb

Open the exact Colab notebook. Predict the 73,728 / 8,768 parameter ratio, verify standard and separable NCHW paths, inspect residual identity gradients, then prove which parameters changed after one optimizer step.

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57.803M $\rightarrow$ 6.874M MACs

PRESERVE

$x + F(x)$ and the direct gradient term

REUSE

512 $\rightarrow$ 6 head: 3,078 trainable

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512 $\rightarrow$ 6 head: 3,078 trainable

The synthetic TinyBackbone verifies freezing mechanics; because it is randomly initialized, it is not evidence that pretrained transfer improves validation.

The transfer decision is small, but the protocol is complete

START

512 $\rightarrow$ 6 head = 3,078 trainable
parameters

SELECT

validation chooses frozen/late/all and
the checkpoint

REPORT

sealed test once; include trainable
count and measured latency

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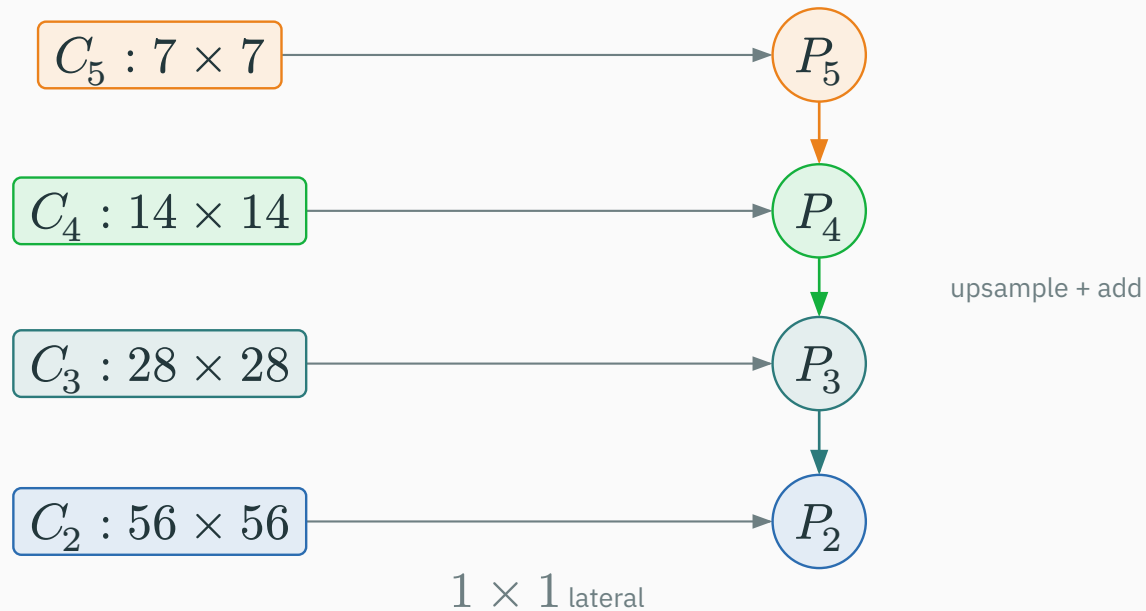
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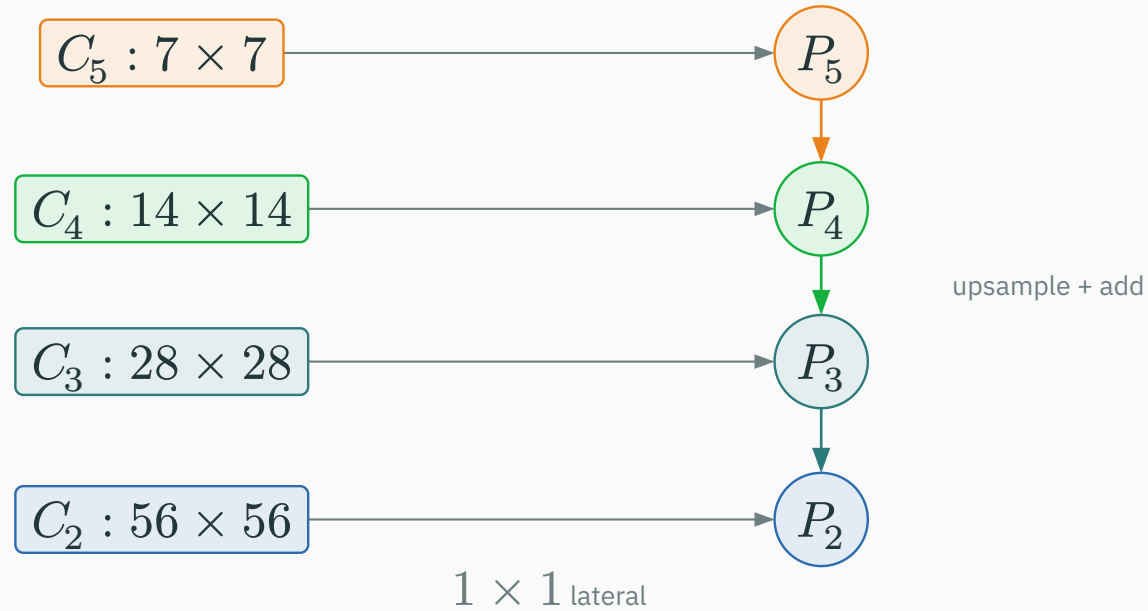
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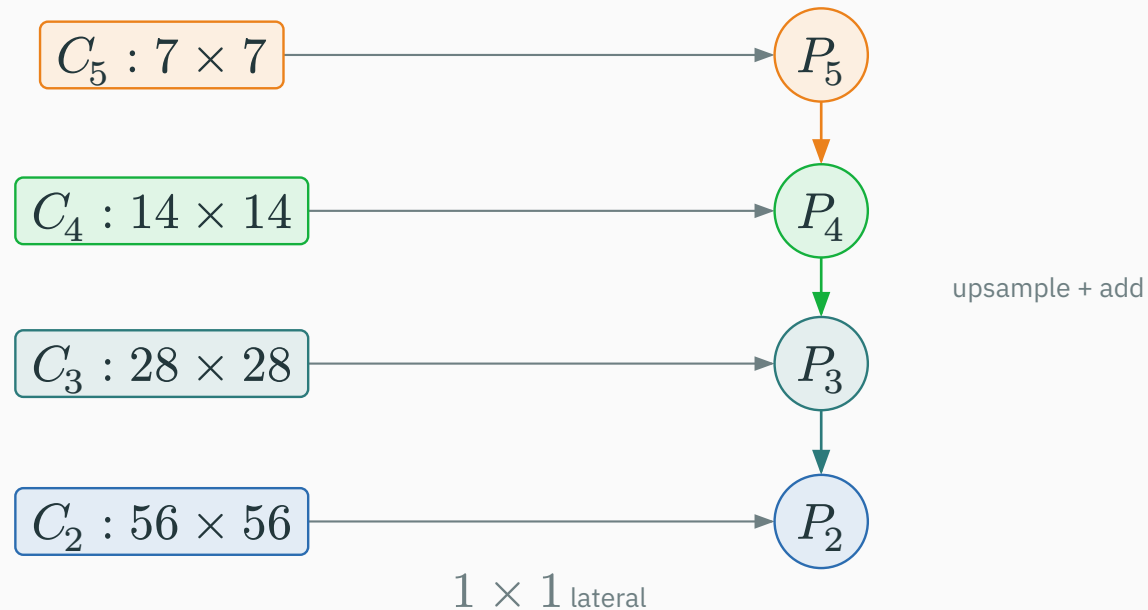
sealed test once; include trainable
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“Transfer learning worked” means a controlled candidate won on validation and survived the one-time test—not merely that training ran.





A feature pyramid laterally projects backbone stages, upsamples coarse features, and adds only tensors with matching height, width, and channels.



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Today the backbone is the reusable object. L10 will derive the localization heads and pyramid operations needed for detection.

need more context → stack small kernels (**VGG**)

need cheaper channel work → squeeze with 1×1 (**bottleneck**)

need several scales → branch then concatenate (**Inception**)

need trainable depth → add identity or projected shortcuts (**ResNet**)

need mobile efficiency → factor spatial and channel work (**MobileNet**)

need a new task → reuse backbone, replace head, validate adaptation

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**Names provide provenance; shape, compute, route, and validation
provide the reasoning.**

Revisit the opening commitment before revealing it

HELD CASE · SAME DEPLOYMENT CONTRACT $28 \times 28 \times 64 \rightarrow 28 \times 28 \times 128$, at least 3×3 context, fewer than 10M convolution MACs.

A · STANDARD

73, 728 weights; 57.803M MACs

B · FACTORIZED

8, 768 weights; 6.874M MACs

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sharing alone guarantees budget

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Defend your original choice using the ledger—not the architecture name.

Answer: B meets every stated contract

A ANSWER

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SHAPE

$28 \times 28 \times 128$ on both paths

CONTEXT

depthwise 3×3 supplies the
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BUDGET

$6.874\text{M} < 10\text{M}$

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$28 \times 28 \times 128$ on both paths

CONTEXT

depthwise 3×3 supplies the
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BUDGET

$6.874\text{M} < 10\text{M}$

Factorization won this contract. A bottleneck, Inception, or projected residual candidate may win when context, branching, or optimization is the dominant need.

Given an image task, ask in order:

1. What output shape and receptive field must the evidence satisfy?
2. Which operation mixes space; which mixes channels?
3. Are additions and concatenations shape-legal?
4. Which weights and output positions dominate MACs?
5. Does measured device latency agree with the ledger?
6. Can a pretrained backbone be probed before broader adaptation?
7. Which validation rule chooses the regime before the test is opened?

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A strong CNN pipeline is a sequence of declared contracts and measured decisions—not a pile of named blocks.

L10 handoff: the backbone answers what; detection must retain where V VISUAL

CLASSIFICATION

$7 \times 7 \times 512 \xrightarrow{\text{global average}} 512 \rightarrow 6$
one image-level answer

→

DETECTION

$C_2, C_3, C_4, C_5 \rightarrow$ spatial heads
classes **and** boxes at many locations

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DETECTION

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classes **and** boxes at many locations

The next lecture keeps the reusable backbone but changes the head, targets, geometry, and evaluation.

BLOCKS

Simonyan & Zisserman · VGG
Szegedy et al. · Inception
He et al. · ResNet
Howard et al. · MobileNet

SCALING + PIPELINES

Tan & Le · EfficientNet
Liu et al. · ConvNeXt
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Use the papers for historical claims; use the held ledger and notebook for reproducible arithmetic.